FIITJEE

ALL INDIA TEST SERIES

PART TEST – I

JEE (Advanced)-2020

PAPER – 1

TEST DATE: 10-11-2019

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section–A (01 – 04, 19 – 22, 37 – 40)*** *contains 12 multiple choice questions which out of 4 options have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*
   Section-A (05 – 12, 23 – 30, 41 – 48) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*

   *Negative Marks : **–1** In all other cases.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

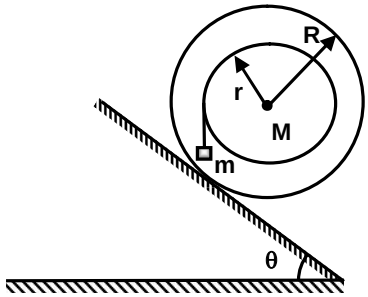
Physics

PART – I

SECTION – A
(One Options Correct Type)

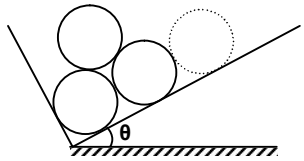
This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A bobbin of mass M = 3 kg consists of a central cylinder of radius r = 5 cm and two end plates of radius R = 6 cm. It is placed on a slotted incline on which it will roll but not slip, and a mass m = 4.5 kg is suspended from a cord wound around the bobbin, as shown in Figure. It is observed that the system is in static equilibrium. What is the angle of tilt θ of the incline?



(A) $\theta = 30°$
(B) $\theta = 45°$
(C) $\theta = 60°$
(D) $\theta = 75°$

2. Smooth, identical logs are piled in a stake truck. The truck is forced off the highway and comes to rest on an even keel lengthwise but with the bed at an angle θ with the horizontal, as shown in Figure. As the truck is unloaded, the removal of the log shown dotted leaves the remaining three in a condition where they are just ready to slide, that is, if θ were any smaller, the logs would fall down. Find θ.



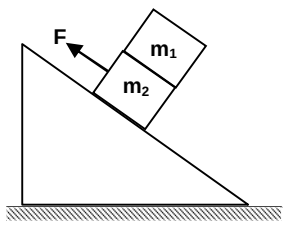
(A) $\theta = \tan^{-1}\left(\frac{1}{2\sqrt{3}}\right)$

(B) $\theta = \tan^{-1}\left(\frac{1}{3\sqrt{3}}\right)$

(C) $\theta = \tan^{-1}\left(\frac{1}{4\sqrt{3}}\right)$

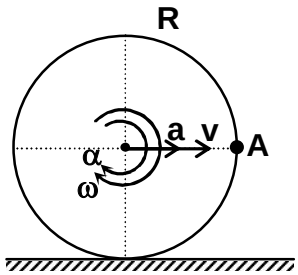
(D) $\theta = \tan^{-1}\left(\frac{1}{5\sqrt{3}}\right)$

3. In the arrangement shown in Figure, the inclined plane is 130 cm long and its upper end is 50 cm above the level of the lower end. The block m_2 rests on the plane, and has a mass of 60 kg. The block m_1 has a mass 200 kg. The coefficient of static friction between the two blocks is 0.50; the coefficient of sliding friction between the lower block and the plane is 0.33. A force F upward and parallel to the plane is applied to the lower block. What is the maximum value of F before the two blocks move with respect to each other?



(A) F = 2000N
(B) F = 2400N
(C) F = 2700N
(D) F = 3600N

4. A uniform circular disc of radius 'R' is rolling without slipping on a rough horizontal surface with a constant acceleration 'a'. Then the radius of curvature of trajectory of point 'A' of the disc relative to the ground at the given instant as shown in the figure is

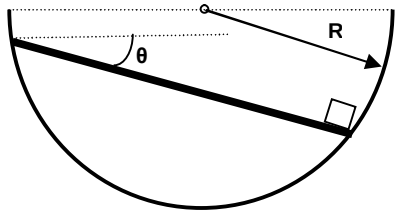


(A) 4R
(B) $2\sqrt{2}\,R$
(C) 2R
(D) $\sqrt{2R}$

(One or More than one correct type)

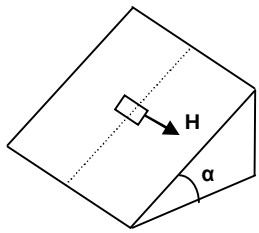
This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

5. A uniform plank of weight W and length $\sqrt{3}R$ lies in a smooth circular trough of radius R , as shown in Figure. At one end of the plank is a weight W/2. Find the possible values of angle θ at which the plank lies when it is in equilibrium.



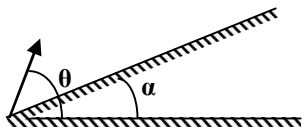
(A) $\theta = 30°$
(B) $\theta = 0°$
(C) $\theta = 25°$
(D) $\theta = 15°$

6. A particle of weight W rests on a rough inclined plane that makes an angle α with the horizontal, as shown in Figure . If the coefficient of static friction $\mu = 2 \tan \alpha$). A horizontal force H is acting transverse to the slope of the plane (see figure)



(A) If H = $2\sqrt{3}$ W sin α the particle will move on the incline plane.
(B) If H = 0.5W sin α the particle will move on the incline plane.
(C) If H = $\sqrt{3}$ W sin α the direction ϕ in which the block goes, relative to H is 30°.
(D) If H = 0.5W sin α the direction ϕ in which the block goes, relative to H is 60°.

7. A perfectly elastic particle is projected with velocity v at an elevation θ. A smooth plane passes through the point of projection and is inclined at an angle α to the horizontal. The particle will return to the point of projection provided cot α cot($\theta - \alpha$) is :

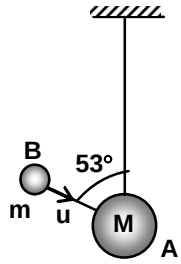


(A) 2
(B) 2.5
(C) 3
(D) 4

8. If a particle of mass m collides elastically with one of mass M at rest, and if the former is scattered at an angle θ and the latter moves at an angle ϕ with respect to the line of motion of the incident particle, then which of the following combinations are possible:
   (A) $\theta = 30°$, $\phi = 0°$, m = 1kg, M = 1kg
   (B) $\theta = 60°$, $\phi = 0°$, m = 3kg, M = 3kg
   (C) $\theta = 45°$, $\phi = 45°$, m = 2kg, M = 2kg
   (D) $\theta = 60°$, $\phi = 45°$, m = 1kg, M = $\sqrt{3}$kg

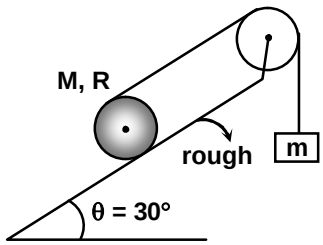
9. A locomotive engine, of mass M=1kg, has two pairs of wheels, of radius a = 0.25m, the moment of inertia of each pair about its axis being Mk^2; and the engine exerts a torque L = $\frac{33}{17}$Nm on the forward axle. If both pairs of wheels commence to roll without sliding when the engine starts and F_μ is the friction between each of the front wheels and the line capable of being called into action. If k = 1, the values of F_μ (in newton) can be:
   (A) 1
   (B) 1.5
   (C) 2.5
   (D) 3

10. A ball 'A' of mass M = 4 kg is suspended by a vertical string. Another ball 'B' of mass m = 1 kg moving with a velocity u = 5.8 m/s at an angle $\theta = 53°$ from vertical collides elastically with the ball 'A' as shown. Then choose the correct option(s).



   (A) The velocity of ball 'A' just after collision is 2 m/s
   (B) The velocity of ball 'B' just after collision is 4.2 m/s
   (C) The velocity of ball 'B' just after collision is 2.5 m/s
   (D) The impulse on the ball 'A' due to tension in the string is 6 N-s

11. A block of mass m = 1 kg is attached to one end of an ideal string whose other end is wound over a solid cylinder of mass M = 8 kg and radius R = 10 cm as shown when the cylinder rolls without slipping down the rough inclined plane of inclination $\theta = 30°$, the block moves vertically upward. Then choose the correct option(s). ($g = 10\ m/s^2$)



   (A) The acceleration of block is 2.5 m/s^2
   (B) The acceleration of block is 5 m/s^2
   (C) The tension in the string is 12.5 N
   (D) The frictional force acting on the cylinder is 17.5 N

12. Consider a right solid cone, whose height is h and the radius of whose base is a.
   (A) The moment of inertia of cone about its slant height is $\frac{3Ma^2\left(6h^2+a^2\right)}{20\left(h^2+a^2\right)}$
   (B) The moment of inertia of cone about an axis passing through its centre of gravity and perpendicular to its axis. $\frac{3M\left(h^2+4a^2\right)}{80}$.

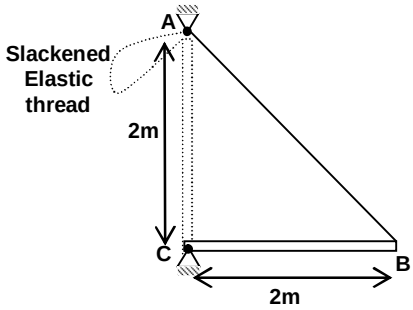
(C) The moment of inertia of cone about its axis is $\frac{3Ma^2}{10}$.

(D) The moment of inertia of cone about any axis passing through the centre of circular base and lying in the plane of the circle is same.

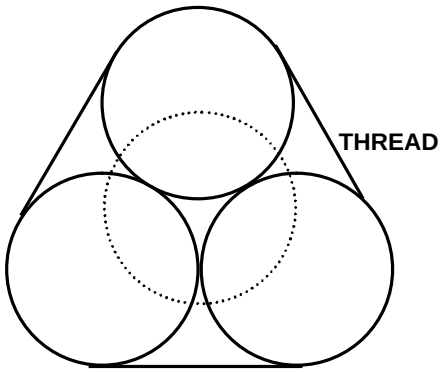
SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

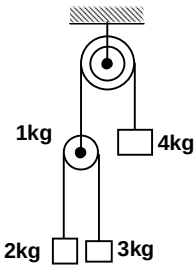
13. A light elastic string of natural length 2m has one end, A, fixed and the other, B, attached to one end of a uniform rod BC of length 2m and mass m = $3-2\sqrt{2}$ kg . This can turn freely in a vertical plane about its other end C, which is fixed at a distance 2m vertically below A. Initially the rod is vertical, and, on being slightly displaced, falls until it is horizontal, and then rises again. Find the elastic constant of string in N/m .(take g = 10m/s^2)



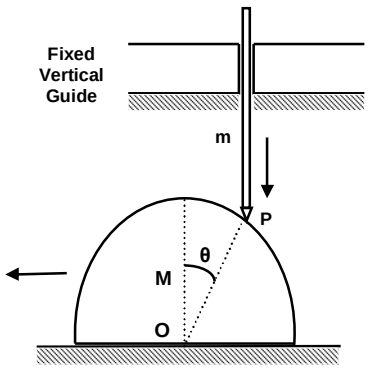
14. Three identical smooth spheres each of mass m = $\frac{3\sqrt{3}}{2}$kg on a smooth horizontal plane are in contact with one another, and are kept together by an endless string in the plane of their centers, just fitting them; when a fourth identical sphere is placed on them, the tension in the string is $\frac{k}{\sqrt{2}}$ newton, then find the value of k. (Take g= 10m/s^2).



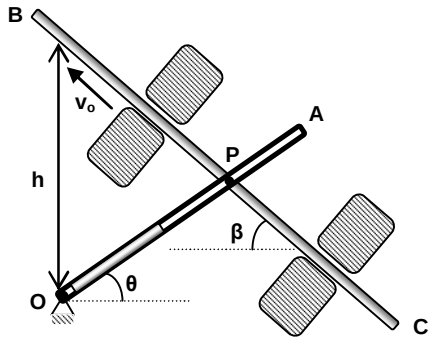
15. Over a smooth light pulley is passed a string supporting at one end a weight of mass 4 kg. At the other end of the string is a pulley of mass 1 kg. Two masses of 2 kg and 3 kg are suspended on either side of the 1kg pulley as shown in figure. The acceleration of the 4 kg mass is ng/49. Find n.



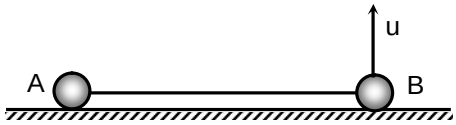
16. A smooth hemisphere, of mass M = 4kg and radius a = 2m, is placed with its plane base on a smooth table. A rod, of mass m = 2kg is constrained to move in a vertical line with one end P on the curved Surface of the hemisphere. Find the angular speed (in rad/s) of point P with respect to centre O of the hemisphere when OP makes an angle $\theta = 60°$ with the vertical. Initially the rod was at the topmost point of the hemisphere.



17. Pin P is attached to BC and slides freely in the slot of OA. Determine the rate of change $d\theta/dt$ (in rad/s) of the angle θ at the instant when h = 4 m, $\beta = 60^0$, and $\theta = 30^0$, knowing that BC moves at a constant speed $v_0 = 5$ m/s.



18. Two identical small balls A and B each of mass m connected by a light inextensible string of length ℓ = 67.5 cm are placed on a smooth horizontal surface. With what minimum velocity u (in m/s) should the ball B be projected vertically upwards so that the ball A leaves the horizontal surface? ($g = 10\ m/s^2$)



Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. In hydrogen atom during de-excitation of electron from an excited state to ground state 15 spectral lines are emitted. Find the longest wavelength of U.V. region during de-excitation.
    (A) $\frac{4}{3R_H}$
    (B) $\frac{36}{35R_H}$
    (C) $\frac{25}{24R_H}$
    (D) $\frac{16}{15R_H}$

20. 0.167 gram of $KBrO_3$ is reacted with excess of KI. And liberated I_2 requires 75 ml of $Na_2S_2O_3$. What is normality of $Na_2S_2O_3$ (where $KBrO_3$ is converted in KBr) (at. wt. of Br is = 80)
    (A) 0.08
    (B) 0.16
    (C) 1.8
    (D) 0.2

21. N_2 and H_2 are 1 : 3 molar ratio, allowed to react to form NH_3, at equilibrium 50% of N_2 and H_2 are reacted. If the total pressure of the system was 21 atm. Find partial pressure of NH_3 at equilibrium.
    (A) 5 atm
    (B) 7 atm
    (C) 9 atm
    (D) 11 atm

22. An alkaline earth metal whose sulphate is soluble and whose oxide is inert on heating. It forms insoluble hydroxide which is soluble in excess of NaOH. Find the metal
    (A) Mg
    (B) Ba
    (C) Ca
    (D) Be

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

23. Which of the following are correct?
    (A) Number of wave made by an electron is two in an orbit where maximum magnetic quantum number is 2.
    (B) Number of electrons in Ne having their orbital angular momentum equal to zero are four.
    (C) Energy of 3d subshell will be more than 3p in hydrogen atom.

(D) The number of degenerated orbitals will be 9 in a orbit of hydrogen atom where energy is proportional to $\left(-\frac{R_H}{9}\right)$.

24. Hydrogen like species in ground state absorb n photons having same energy and during de-excitation emits exactly n photons. Then the energy of absorb photon would be
(A) 40.8 eV
(B) 91.8 eV
(C) 163.2 eV
(D) 57.2 eV

25. Which of the following is/are true regarding H_2 gas
(A) H_2 is more rapidly adsorbed on to the surface than D_2.
(B) H_2 reacts with Cl_2 13 times faster than D_2 with Cl_2.
(C) NaCl is more soluble in D_2O than H_2O.
(D) H_2 can be obtained from water gas by oxidation of CO into CO_2 which can be easily removed by dissolving in H_2O.

26. Which are true regarding H_2O_2?
(A) Decomposition of H_2O_2 can be catalysed by addition of MnO_2.
(B) H_2O_2 can be prepared by adding TiO_2 in dil. H_2SO_4.
(C) H_2O_2 can be prepared by adding H_3PO_4 in hydrated BaO_2.
(D) H_2O_2 decomposes on standing and on heating this can be retarded by using acetanilide or stannates.

27. Select the correct statement:
(A) F-Centre is responsible for imparting the colour to the crystal.
(B) Ionic compound having low co-ordination number shows Frenkel defect.
(C) Schottky defect is more common for crystal having high radius-ratio (cation to anion).
(D) Stoichiometry of crystal remains unaffected due to Schottky defect.

28. Select the CORRECT statements:
(A) Mixture of NaH_2PO_4 and Na_2HPO_4 act as a buffer solution.
(B) pH of a solution formed by mixing equimolar quantities of HCOOH and HCl will be less than that of a solution formed from equimolar quantities of HCOOH and HCOONa.
(C) When Na_3PO_4(aq) is titrated with HCl(aq), the pH of solution at second equivalent point is $\frac{1}{2}\left(pK_{a_1} + pK_{a_2}\right)$
(D) Phenolphthalein changes it's colour from pink to colourless at first equivalence point of titration of Na_2CO_3 with HCl.

29. Heating of lime stone given below

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2 \uparrow$$

Equilibrium will shift in forward direction.
(A) addition of He gas at constant pressure
(B) By increasing the temperature
(C) Decreasing the volume of the vessel
(D) Addition of CaO

30. Which of the following statements regarding alkali metal are correct?
(A) When alkali metal dissolve in liquid NH_3, as conc. of NH_3 increases paramagnetic character decreases.
(B) Alkali metal in liquid NH_3 shows very strong oxidizing character.
(C) Stability of superoxide of alkali metals increases with increase in size of metal cation.
(D) All alkali metal carbonates decomposes on heating and gives CO_2 gas.

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. A unit cell has one atom on each corner of cube and two atoms at one of it's body diagonals if volume of unit cell is 5×10^{-24} cm^3 and density of element is 15 gram/cm^3. The number of atom present in 150 gram of element is $x \times 10^{24}$ find x. (Avogadro number = 6×10^{23})

32. $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$
If at equilibrium some CO_2 at 1 atm is added at constant volume, 25% of original NH_3 and CO_2, solidified before equilibrium was re-established. Find total pressure at new equilibrium.

33. $3C_2H_2 \xrightarrow[\text{Cu tube}]{\text{Red hot}} C_6H_6$
Find the degree of association of ethyne if observed molecular mass of mixture is 39.

34. $A(s) \rightleftharpoons B(s) + 2C(g)$
K_P of above reaction is 9 atm^2, at 327°C
A 10 litre vessel contains 1 mole of B. How many moles of C would be added to drive backward reaction to completion.

35. 50 ml of 0.1 M Na_3PO_4 was titrated with 50 ml of 0.2 M HCl. Find the pH of resulting mixture where pK_{a_1}, pK_{a_2} and pK_{a_3} of H_3PO_4 are 5, 8, 13.

36. A single electron species having atomic number Z make a transition from 2 to 1. The emitted photon is absorbed by excited electron of Li^{+2} in n_1 orbit and further excited to n_2. Find the value of $\frac{2n_1 + 3n_2 + 6Z}{10}$

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. If $\int_0^1 \frac{\sin(\ln x)}{\ln x} dx$ is equal to

(A) $\frac{\pi}{4}$

(B) $\frac{11}{14}$

(C) $\frac{\pi}{4} - \frac{11}{14}$

(D) $\frac{22}{7} - \pi$

38. Number of continuous function(s) f: [0, 1] $\to$ R satisfying $\int_0^1 f(x)dx = \frac{1}{3} + \int_0^1 f^2(x^2)dx$ is/are

(A) 1
(B) 2
(C) 3
(D) 4

39. Let f, g : R $\to$ [1, ∞) are two differentiable function on the real line satisfying the differential equation $(f^2 + g^2)f' + (fg)g' = 0$, then
(A) f is bounded, but g not
(B) f is unbounded, but g is bounded
(C) both are unbounded
(D) both f and f·g are bounded

40. The area of the set of points (x, y) in the plane that satisfy that two inequalities $x^2 + y^2 \le 2$ and $x^4 + x^3y^3 \le xy + y^4$ is
(A) π
(B) $\frac{2\pi}{3}$
(C) $\frac{19\pi}{21}$
(D) none of these

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

41. Let f(x) be a real polynomial of degree 4 whose graph has two real inflection points. There are three regions bounded by the graph and the line passing through these inflection points, then
(A) Two of these regions have equal area

(B) Area of one of the region is equal to the sum of other two
(C) Area of one of the region is double the sum of areas of other two
(D) Area of one of the region is square of the sum of areas of other two

42. Let a curve passing through (1, 0) satisfies the differential equation
$\left(2xe^{y}+3y^{2}\right)\frac{dy}{dx}+\left(3x^{2}+2e^{y}\right)=0$, then
(A) $x^3 + 2xe^y + y^3 = 3$
(B) $x^3 - 2xe^{2y} + y^3 = -1$
(C) The slope of tangent to the curve at (1, 0) is $-\frac{5}{2}$
(D) The curve intersects x-axis at only one point

43. Let f be a function having a continuous derivative on [0, 1] and with the property that $0 < f'(x) \le 1$. Also, suppose that f(0) = 0, then $\frac{\left[\int_0^1 f(x)dx\right]^2}{\int_0^1 (f(x))^3 dx}$ is always greater than or equal to
(A) $\frac{2}{3}$
(B) $\frac{1}{2}$
(C) $\frac{1}{3}$
(D) 1

44. Let S be the set of points in the Cartesian plane that satisfy $|||x| - 2| - 1| + |||y| - 2| - 1| = 1$, if a model of S were built from wire of negligible thickness, then
(A) total length of wire required would be $64\sqrt{2}$
(B) total length of wire required would be $32\sqrt{2}$
(C) area enclosed by the wire is 50 sq. units
(D) area enclosed by the wire is 25 sq. units

45. Which of the following are CORRECT? (where [.] denotes the greatest integer function and {.} fractional part of x)
(A) Right hand derivative of $f(x) = |\sin x| \cos^{-1} \cos x$ at non-differentiable point is π
(B) Left hand derivative of $f(x) = [x] |\sin \pi x|$ at non-differentiable point is $(1 - k)\pi$, $k \in I$
(C) Left hand derivative of $f(x) = [x] \sin^2 \pi x$ at non- differentiable point is $(-1)^k\pi$
(D) Right hand derivative of $f(x) = |x - 1|\{x\} : x \in \left[\frac{1}{2}, \frac{3}{2}\right]$ at non- differentiable point is 0

46. The function f is defined by y = f(x) where $x = r(3 \cos t + 2|\cos t|)$
and $y = r(2 \sin t + |\sin t|) : t \in [0, \pi]$, then
(A) f(x) is continuous on [–r, 5r]
(B) f(x) is discontinuous at one point on [–r, 5r]
(C) f(x) is differentiable $\forall\ x \in$ [–r, 5r]
(D) f(x) is not differentiation at 2 points on [–r, 5r]

47. $\int \frac{x^4+1}{x^6+1}dx$ is

(A) $\tan^{-1}x + \frac{1}{3}\tan^{-1}(x^3) + c$

(B) $\frac{1}{3}\tan^{-1}\left(\frac{3x-3x^5}{1-3x^2-3x^4+x^6}\right) + c$

(C) $\frac{1}{3}\tan^{-1}\left(\frac{2x-x^4}{1-2x^5-3x^2}\right) + c$

(D) $\frac{1}{2}\tan^{-1}\left(\frac{x-x^3-1}{1+x^2-3x^5}\right) + c$

48. Let f: [0, 1] $\to$ R be a double differentiable function with f(0) = f(1) = 0, and f″(x) + 2f′(x) + f(x) $\geq$ 0 $\forall$ x $\in$ (0, 1), then

(A) f(x) $\leq$ 0 $\forall$ x $\in$ [0, 1]

(B) if e^x f(x) assumes its minimum value in [0, 1] at $x = \frac{1}{\sqrt{2}}$, then f(x) + f′(x) < 0 $\forall$ $x \in \left(0, \frac{1}{\sqrt{2}}\right)$

(C) if g(x) = e^x f(x), then number of real solution of the equation g(g(g(x))) = 0 is 2

(D) f(x) > 0 $\forall$ x $\in$ [0, 1]

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. Number of real solution(s) of the equation $6^x + 1 = 8^x - 27^{x-1}$ is equal to

50. Let A be the minimum value of the function f: R $\to$ R, $f(x) = \frac{(x^2-x+1)^3}{x^6-x^3+1}$, then $\frac{2+\sqrt{3}}{5\sqrt{3}}$A is

51. If the equation $x^4 + ax^3 + 2x^2 + bx + 1 = 0$; (a, b $\in$ R) has a real solution, then minimum value of $a^2 + b^2$ is

52. $\lim_{n\to\infty} \frac{1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{n^7}}{\ln n}$ is equal to

53. Let y = f(x) be a one-one function satisfying the differential equation $\frac{d^2y}{dx^2} + \frac{d^2x}{dy^2} = 0$ and f′(0) = 1, f(1) = 2, then f(0) is

54. Number of continuously differentiable function f: R $\to$ R satisfying f(x) > 0 and f′(x) = f(f(x)) $\forall$ x $\in$ R is/are